

## SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

### Sub-Strand 3.1: Statistics 1 — Teacher Reference (Pre-filled)

| SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10 |                                                                                                                                           |
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| Sub-Strand                                    | Sub-Strand 3.1: Statistics 1                                                                                                              |
| Driving Question                              | DRIVING QUESTION: How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                          | What did I observe?                                                                                                                                                                                                                                                                                                                                        | What did I learn?                                                                                                                                                                                                                                                                                                                              | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>Lesson 1</b><br><b>Whose Claim Is Right?</b><br><b>Launching the Principal's Problem</b> | What did we observe? We collected 40 responses from Grade 10 Green about main travel mode (Walk, Matatu, Boda boda, Bicycle, Private car, School bus), saw three conflicting claims from Atieno, Brian and Zawadi, and found the long raw list was hard to use to say who is right.                                                                        | What did we learn? Raw data is an unorganised list; a claim needs evidence; a long list does not settle an argument by itself, and one class is not the whole school.                                                                                                                                                                          | How can we explain it so far? We have data but not yet evidence for Mr Otieno because the list is unorganised. We predict that organising and showing the data will make the answer clearer, but we still need to find out how, and which summary or graph to trust.                                                                                                                                                                                                                                                                                     |
| <b>Lesson 2</b><br><b>Tally and Frequency Tables: Making Sense of the 40 Responses</b>      | We tallied the 40 raw responses from Grade 10 Green and got Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4, which add to 40. Percentages are Walk 35%, Matatu 30%, Boda boda 10%, Bicycle 7.5%, Private car 7.5%, School bus 10%. For the 30 trees in the Lakeview compound we got Grevillea 12, Cypress 8, Eucalyptus 6, Neem 4. | A frequency table has the columns Category (or Value), Tally and Frequency. Tally marks are grouped in fives, and the frequencies must add up to the total number of responses. $\text{Percentage} = f / n \times 100$ . Categorical data (travel mode, tree type) and discrete data (counts such as siblings) can both be organised this way. | The raw list could not settle the argument, but the table shows Walk (14 of 40, 35%) is the largest group and Matatu (12 of 40, 30%) is second. Neither is more than half, so 'almost everyone walks' and 'most come by matatu' are both wrong, although together they make 26 of 40, which is 65%. Only 4 of 40 use the school bus and 3 of 40 cycle. The table cannot settle Zawadi's claim about the single 14-seater school bus, and one class is a small sample, so we need typical values, graphs and whole-school data before advising Mr Otieno. |
| <b>Lesson 3</b><br><b>Typical Value: Mode and Mean</b>                                      | The travel table has its highest frequency at Walk (14 of 40, 35%). The siblings table has its highest frequency at 2 siblings (12 of 40). Adding an fx column gives $\Sigma fx = 83$ for                                                                                                                                                                  | The mode is the value or category with the highest frequency and works for categories and numbers. The mean = $\Sigma fx \div \Sigma f$ and works only for numbers. Mean of siblings =                                                                                                                                                         | The typical Grade 10 Green learner walks to school and has about 2 siblings. The mean (2.1) need not be a whole number because it is a fair share. These summaries                                                                                                                                                                                                                                                                                                                                                                                       |

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|                                                                                      | $\Sigma f = 40$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $83 \div 40 = 2.075$ , about 2.1 siblings; mode = 2. Travel mode: Walk is the mode but is not a majority.                                                                                                                                                                                                          | describe only this one class, so they are not enough alone to support a decision by Mr Otieno; we need to know when each measure is the best choice (Lesson 4).                                                                                                                                                                                                                                                        |
| <b>Lesson 4</b><br><b>Median and Choosing the Best Measure</b>                       | We ordered the siblings data and found the 20th and 21st values both equal 2, so the median is 2 siblings (mean 2.1, mode 2). We ordered the 7 Eldoret club 800 m times and found the median 106 s, the mode 105 s and the mean 109.3 s; without the 130 s time the mean of the other six is 105.8 s.                                                                                                                                                                                   | The median is the middle value of ordered data (for odd n the $((n+1)/2)$ th value; for even n the mean of the two middle values). One very large or small value (an outlier) pulls the mean but hardly moves the median. For categories such as travel mode only the mode (Walk, 14 of 40) can be used.           | The mean, median and mode agree for the siblings data because it is evenly spread, but they differ for the runner times because one slow time (130 s) pulls the mean up. So the best summary depends on the type of data and the question. For Mr Otieno this means choosing and explaining the measure honestly, and remembering that these data are from only one class.                                             |
| <b>Lesson 5</b><br><b>Bar graphs and line graphs: which picture fits which data?</b> | Bar graph of travel: Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4 (total 40), with the tallest bar for Walk. Line graph of Mama Achieng's mandazi sales: Mon 40, Tue 45, Wed 42, Thu 50, Fri 65, Sat 30, Sun 20, peaking on Friday and falling to Sunday.                                                                                                                                                                                                    | A bar graph suits categories, with separate equal-width bars and a frequency axis starting at 0. A line graph suits change over time, with time on the horizontal axis and points joined by straight lines. Both need labelled axes and a title.                                                                   | The same table of travel data becomes clear evidence in a bar graph: Walk is 14 of 40 (35%) and Matatu is 12 of 40 (30%), so neither claim shows a majority, though together they are 65%. Bicycle (3 of 40) gives little support for a shed. We should choose the graph that matches the data type, and collect whole-school data before deciding on the bus or lunch serving.                                        |
| <b>Lesson 6</b><br><b>Pie charts: turning frequencies into angles</b>                | We observed that the 40 travel responses can be shown as a pie chart: Walk 126 degrees, Matatu 108 degrees, Boda boda 36 degrees, Bicycle 27 degrees, Private car 27 degrees and School bus 36 degrees, which add up to 360 degrees. Walk takes 14 of 40 learners (35%) and Matatu 12 of 40 (30%), so together 65%, but neither is more than half the circle. We also drew the Wanjiru family budget of KSh 30,000 (Food 144, Rent 90, School fees 54, Transport 36, Savings 36 = 360). | We learned that $\text{angle} = f / n \times 360$ , that each learner in our class is worth 9 degrees, that the angles of a pie chart must add up to 360, and that a pie chart shows each part as a share of the whole while a bar graph shows exact counts and is better for comparing close values.              | We can explain that the pie chart helps Mr Otieno see at a glance that walking and matatu are the biggest groups but neither is the majority, and that only 3 of 40 (7.5%) cycle, so this class gives little support for a bicycle shed. Because this is only one class, we advise collecting the same data from the whole school before deciding on the bus or second lunch serving.                                  |
| <b>Lesson 7</b><br><b>Reading Graphs Critically and Using Digital Tools</b>          | We observed that the same travel data looked very different on two bar graphs: with the axis starting at 0, Walk (14) is about 4.7 times Bicycle (3); with the axis starting at 2, Walk looked 12 times Bicycle. We also read the pie chart and the mandazi line graph, and recreated the travel graphs on a spreadsheet or app.                                                                                                                                                        | We learned to read exact values from bar graphs, pie charts and line graphs, to support every conclusion with a number, and to check a graph for a frequency axis that starts at 0, equal steps, labels, a title and pie angles that add to 360. We also learned that one class cannot speak for the whole school. | A misleading graph changes how the data looks, not the data itself. Honest graphs of the travel data show Walk 14 of 40 (35%) and Matatu 12 of 40 (30%), together 65%, and only 3 of 40 (7.5%) cycling, so the class gives little support for a bicycle shed. Mr Otieno should trust a graph only after checking its axis and scale, and should collect whole-school data before deciding on the bus or lunch serving. |

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| <b>Lesson 8</b><br><b>Project: Evidence for the Principal</b> | What did you observe? Write what the raw list of 40 responses looked like, and what the frequency table, bar graph and pie chart for Grade 10 Green showed (Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4), including the mean, median and mode of the runner times. | What did you learn? Explain how organising, summarising and displaying data turns a messy list into evidence, why neither 'almost everyone walks' nor 'most come by matatu' is right, and why the best summary or graph depends on the data and the question. | How can you explain the answer to the driving question? Write your Final Explanation for Mr Otieno: what the evidence says about the bicycle shed, school bus and lunch serving, and why he should collect whole-school data before deciding. |
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## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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